

A Log-Linear Regression Analysis of Allometric Scaling and Species Effects on Penguin Body Mass

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Abstract

The latent-grade model uses the trimmed file `nodules_train_trimmed.csv` rather than the entire training data: we removed individual ratings at least two grades from their nodule's median (212 of the 4647 training ratings, 4.6%, across 204 nodules, one of which lost all of its ratings). The naive models and the test set use all ratings. This rule was chosen after an initial fit gave implausible results. It improves the interpretability of the latent classes, but it reduces measured disagreement, so the estimated confusion matrix overstates reader reliability and predictive fit on the untrimmed ratings is worse.

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1 Introduction

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2 Data Description

The dataset used in this study is the LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) database[1]. It contains thoracic CT scans in which up to four radiologists outlined each lung nodule of at least 3 mm and rated its appearance and its likely malignancy.

The unit of analysis is the *nodule*. Geometric features are averaged across a nodule's readers and semantic ratings are summarised by their median. We use the median because the ratings are ordinal, so the gaps between grades have no fixed numerical meaning, and because it is less sensitive to a single outlying reader than the mean.

The variables used in this study are as follows.

Response

- **Malignancy (`mal_median`)**: The median of the readers' ratings, on an ordered scale from 1 (highly unlikely to be malignant) to 5 (highly suspicious). A median falling between two grades is rounded up, which affects 548 nodules (20.8%). Each reader's rating is also retained for the latent-grade model. Discrete (ordinal), with support $\{1, 2, 3, 4, 5\}$.

Primary model: geometric features These are computed from the outlines drawn by the radiologists and are treated as continuous random variables.

- **Log Volume (`log_volume`)**: The natural logarithm of the nodule's volume in mm^3. The logarithm is used because volume is strongly right-skewed. Continuous, with support $\mathbb{R}$.
- **Elongation (`elongation`)**: A value near 1 means that the nodule is not elongated. Continuous, with support $[0, 1]$.
- **Flatness (`flatness`)**: A value near 1 means that the nodule is not flat. Continuous, with support $[0, 1]$ and a point mass at 0.
- **Solidity (`solidity`)**: The number of voxels in the nodule divided by the number in its convex hull. A low value indicates an irregular outline. Continuous, with

support $(0, 1]$.

- **Compactness (`compactness`)**: $V^{2/3}/S$, for volume V and surface area S. A high value means little surface area relative to volume. Continuous, with support $(0, \infty)$.

Secondary model: primary features and semantic ratings These are ordinal ratings from 1 to 5, given by the same radiologists who rate malignancy. The secondary model adds them to the five geometric features above. Since each is a median across readers, all five are discrete (ordinal), with support $\{1, 1.5, 2, \ldots, 4.5, 5\}$.

- **Spiculation (`spiculation`)**: Spiky extensions from the edge, from none to marked.
- **Lobulation (`lobulation`)**: Rounded bulges along the edge, from none to marked.
- **Margin (`margin`)**: Definition of the edge, from poorly defined to sharp.
- **Texture (`texture`)**: Internal density, from non-solid to solid.
- **Sphericity (`sphericity`)**: Overall roundness, from linear to round.

The data are split by patient into a training set (609 patients, 1809 nodules) and a test set (261 patients, 828 nodules). Flatness is exactly 0 for 45 training nodules whose outlines lie in a single plane.

2.1 Summary Statistics

All summary statistics are computed on the training set. Table 1 gives the unstandardised values of the predictors, and Table 2 breaks them down by median malignancy grade.

Model	Variable	Mean	Minimum	Maximum	Std. Dev.
Primary	Log Volume	5.118	2.442	10.345	1.394
	Elongation	0.753	0.306	0.988	0.121
	Flatness	0.555	0.000	0.921	0.164
	Solidity	0.795	0.228	1.000	0.106
	Compactness	0.177	0.058	0.312	0.044
Secondary	Spiculation	1.476	1.000	5.000	0.889
	Lobulation	1.572	1.000	5.000	0.883
	Margin	3.899	1.000	5.000	1.138
	Texture	4.363	1.000	5.000	1.202
	Sphericity	3.789	1.000	5.000	0.826

Table 1: Overall Summary Statistics ($n = 1809$ training nodules)

Model	Variable	Grade 1 (233)	Grade 2 (378)	Grade 3 (823)	Grade 4 (239)	Grade 5 (136)
Primary	**Log Volume**	4.563 (0.981)	4.474 (0.935)	4.779 (0.983)	6.349 (1.269)	7.744 (1.082)
	Elongation	0.749 (0.135)	0.731 (0.125)	0.758 (0.117)	0.773 (0.115)	0.761 (0.109)
	Flatness	0.591 (0.150)	0.511 (0.173)	0.548 (0.167)	0.584 (0.151)	0.602 (0.132)
	Solidity	0.842 (0.092)	0.812 (0.092)	0.808 (0.093)	0.742 (0.104)	0.676 (0.125)
	Compactness	0.184 (0.049)	0.187 (0.043)	0.183 (0.040)	0.154 (0.037)	0.135 (0.031)
Secondary	**Spiculation**	1.172 (0.645)	1.200 (0.632)	1.394 (0.766)	1.795 (0.930)	2.699 (1.243)
	Lobulation	1.255 (0.700)	1.300 (0.728)	1.509 (0.819)	1.933 (0.878)	2.618 (0.970)
	Margin	4.674 (0.793)	3.971 (1.148)	3.756 (1.159)	3.669 (1.076)	3.643 (1.011)
	Texture	4.880 (0.560)	4.344 (1.253)	4.197 (1.323)	4.324 (1.139)	4.603 (0.893)
	Sphericity	4.043 (0.924)	3.810 (0.877)	3.779 (0.781)	3.653 (0.790)	3.599 (0.735)

Table 2: Mean (Std. Dev.) by Median Malignancy Grade (number of training nodules in brackets)

2.2 Data Source

The dataset used in this study is the LIDC-IDRI database[1], hosted by The Cancer Imaging Archive[2] at LIDC-IDRI (TCIA). The full image collection is about 133 GB, but the radiologists' annotations are bundled, already parsed, with the Python library `pylidc`[4]. Every quantity used in this project is computed from these annotations with `pylidc` version 0.2.3, so no CT images were downloaded.

3 Model Specification

3.1 Logistic Regression Model (Naive Proportional Odds)

Methodology Let $Y_i \in \{1, \dots, 5\}$ be the median malignancy grade of nodule i, and let x_i be its vector of predictors. Since Y_i is ordered, we model its cumulative probabilities with a proportional odds model (an ordinal logistic regression):

$$\operatorname{logit} P(Y_i \le k \mid x_i) = \alpha_k - x_i^{\mathsf{T}} \beta, \qquad k = 1, \dots, 4,$$

where $\alpha_1 < \cdots < \alpha_4$ are thresholds. A positive β_j means that a larger x_{ij} shifts a nodule towards higher grades. The odds of being above any given grade are multiplied by e^{β_j} for each one-unit increase in x_{ij}, and this factor is assumed to be the same at every threshold. The model is *naive* because it takes the median as the true grade and ignores disagreement between readers.

Writing $F(t) = e^t/(1 + e^t)$ for the logistic distribution function, with $\alpha_0 = -\infty$ and

$\alpha_5 = \infty$, and treating nodules as independent, the likelihood is

$$L(\alpha, \beta) = \prod_{i=1}^{n} \prod_{k=1}^{5} \left[F(\alpha_k - x_i^\top \beta) - F(\alpha_{k-1} - x_i^\top \beta) \right]^{\mathbf{1}\{Y_i = k\}}.$$

The models are fitted by maximising L on the training set with `polr`. All predictors are standardised, so one unit is one standard deviation. Confidence intervals for the odds ratios are profile-likelihood intervals, obtained by inverting the likelihood ratio test of $H_0 \colon \beta_j = b$ using the χ^2_1 limit given by Wilks' theorem, and then exponentiating the endpoints.

Wilks' theorem states that, under regularity conditions and if H_0 is true, the likelihood ratio statistic satisfies

$$\Lambda = 2 \Big[\sup_{\theta \in \Theta} \ell(\theta) - \sup_{\theta \in \Theta_0} \ell(\theta) \Big] \xrightarrow{d} \chi^2_r$$

as $n \to \infty$, where ℓ is the log-likelihood and r is the number of parameters fixed by H_0. This is a large-sample result. Here $r = 1$, so the approximate 95% interval for β_j is the set of values b with $\Lambda(b) \le \chi^2_{1,0.95} = 3.841$.

Primary Model The fitted model is

$$\begin{aligned} \operatorname{logit} \hat{P}(Y_i \le k \mid x_i) = \hat{\alpha}_k - \big(& 1.330 \cdot \texttt{log_volume} + 0.333 \cdot \texttt{elongation} \\ & - 0.173 \cdot \texttt{flatness} - 0.342 \cdot \texttt{solidity} \\ & + 0.047 \cdot \texttt{compactness} \big), \end{aligned}$$

with $\hat{\alpha} = (-2.503, -1.108, 1.674, 3.488)$ and maximised log-likelihood -2155.134. The estimates are given in Table 3.

Predictor	$\hat{\beta}$	Std. Error	Odds Ratio	95% CI
`log_volume`	1.330	0.065	3.781	(3.335, 4.296)
`elongation`	0.333	0.055	1.395	(1.253, 1.555)
`flatness`	-0.173	0.053	0.841	(0.757, 0.934)
`solidity`	-0.342	0.065	0.710	(0.625, 0.806)
`compactness`	0.047	0.059	1.048	(0.934, 1.175)

Table 3: Primary Proportional Odds Model

Interpretation Each odds ratio is for a one standard deviation increase in that predictor, holding the others constant.

- **Size:** Log volume has by far the largest effect. One standard deviation, roughly

a fourfold increase in volume, multiplies the odds of a higher malignancy grade by 3.78.

- **Solidity:** A more solid nodule has lower odds of a higher grade (odds ratio 0.71), so irregular outlines are rated as more suspicious.
- **Elongation and Flatness:** Elongation is close to 1 for a nodule that is not elongated, so an odds ratio of 1.40 means that more elongated nodules have lower odds of a higher grade. Flatness works in the other direction: its odds ratio of 0.84 means that flatter nodules have higher odds.
- **Compactness:** The confidence interval contains 1, so the data are compatible with no effect once the other predictors are included.

Secondary Model The geometric features only describe the outline of a nodule, whereas radiologists also judge its appearance on the scan. We therefore add the semantic ratings to measure how much this assessment explains beyond geometry: spiculation, lobulation and margin describe the edge in terms that radiologists themselves use, texture records internal density, which the outlines cannot capture, and sphericity gives a reader's view of overall shape. Adding these ratings gives

$$\begin{aligned}\operatorname{logit} \hat{P}(Y_i \le k \mid x_i) = \hat{\alpha}_k - \big(&1.210 \cdot \texttt{log_volume} + 0.321 \cdot \texttt{elongation} \\ &- 0.065 \cdot \texttt{flatness} - 0.160 \cdot \texttt{solidity} \\ &+ 0.002 \cdot \texttt{compactness} + 0.346 \cdot \texttt{spiculation} \\ &+ 0.091 \cdot \texttt{lobulation} - 0.280 \cdot \texttt{margin} \\ &+ 0.063 \cdot \texttt{texture} + 0.002 \cdot \texttt{sphericity}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.572, -1.144, 1.735, 3.617)$ and maximised log-likelihood -2108.642. The estimates are given in Table 4.

Predictor	$\hat{\beta}$	Std. Error	Odds Ratio	95% CI
log_volume	1.210	0.067	3.352	(2.942, 3.828)
elongation	0.321	0.056	1.379	(1.237, 1.539)
flatness	-0.065	0.056	0.937	(0.840, 1.045)
solidity	-0.160	0.069	0.852	(0.744, 0.976)
compactness	0.002	0.059	1.002	(0.893, 1.126)
spiculation	0.346	0.067	1.413	(1.241, 1.613)
lobulation	0.091	0.066	1.096	(0.964, 1.246)
margin	-0.280	0.074	0.756	(0.654, 0.874)
texture	0.063	0.067	1.065	(0.933, 1.215)
sphericity	0.002	0.051	1.002	(0.907, 1.107)

Table 4: Secondary Proportional Odds Model

Interpretation

- **Semantic Ratings:** More spiculated nodules have higher odds of a higher grade (odds ratio 1.41), and nodules with sharper margins have lower odds (0.76). The confidence intervals for lobulation, texture and sphericity contain 1.
- **Geometric Features:** Size remains the largest effect (odds ratio 3.35), and elongation barely changes (1.38). The solidity effect moves towards 1 (0.85), and the interval for flatness now contains 1, which suggests that the ratings capture some of the same information about shape.
- **Circularity:** The semantic ratings come from the same radiologists who rate malignancy, so these associations may partly reflect a reader's overall impression of a nodule.

3.2 Latent True-Grade Model (EM Algorithm)

Methodology The naive model treats the median rating as the true grade. Instead, we now treat the true grade $Z_i \in \{1, \ldots, 5\}$ of nodule i as an unobserved variable, and the readers' ratings R_{ij} as noisy measurements of it. The model has two parts:

- **Structural Model:** The proportional odds model now describes the latent grade, using the primary features x_i:

$$p_{ik}(\alpha, \beta) = P(Z_i = k \mid x_i) = F(\alpha_k - x_i^\mathsf{T}\beta) - F(\alpha_{k-1} - x_i^\mathsf{T}\beta).$$

- **Measurement Model:** A reader rates a nodule of true grade k as r with probabil-

ity $P(R_{ij} = r \mid Z_i = k) = \pi_{kr}$, where $\sum_r \pi_{kr} = 1$. All readers share this 5×5 matrix, because reader identities are not consistent across scans. Ratings are assumed to be conditionally independent given Z_i.

Let $\Psi = (\alpha, \beta, \pi)$, and let $C_{ir} = \#\{j : R_{ij} = r\}$ be the number of readers who gave nodule i the rating r. The ratings are the observed (*incomplete*) data, and their log-likelihood is

$$\ell(\Psi) = \sum_{i=1}^{n} \log \sum_{k=1}^{5} p_{ik}(\alpha, \beta) \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

The sum inside the logarithm makes this hard to maximise directly. If Z_i were observed, the *complete* log-likelihood would separate into simpler terms:

$$\ell_c(\Psi) = \sum_{i=1}^{n} \sum_{k=1}^{5} \mathbf{1}\{Z_i = k\} \Big[\log p_{ik}(\alpha, \beta) + \sum_{r=1}^{5} C_{ir} \log \pi_{kr} \Big].$$

We therefore estimate Ψ with the EM algorithm[3].

Core Steps Starting from an initial guess $\Psi^{(0)}$, each iteration $t = 0, 1, 2, \ldots$ does the following.

1. **E-step:** Compute the posterior probability of each true grade given the ratings and the current parameters:

$$w_{ik}^{(t)} = P\big(Z_i = k \mid R_i, x_i, \Psi^{(t)}\big) = \frac{p_{ik}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{kr}^{(t)}\big)^{C_{ir}}}{\sum_{l=1}^{5} p_{il}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{lr}^{(t)}\big)^{C_{ir}}}.$$

   Taking the expectation of ℓ_c then replaces $\mathbf{1}\{Z_i = k\}$ with $w_{ik}^{(t)}$, which gives the Q-function:

$$Q\big(\Psi \mid \Psi^{(t)}\big) = \sum_{i=1}^{n} \sum_{k=1}^{5} w_{ik}^{(t)} \log p_{ik}(\alpha, \beta) + \sum_{i=1}^{n} \sum_{k=1}^{5} \sum_{r=1}^{5} w_{ik}^{(t)} C_{ir} \log \pi_{kr}.$$

2. **M-step:** Set $\Psi^{(t+1)} = \operatorname{argmax}_{\Psi} Q(\Psi \mid \Psi^{(t)})$. The two terms of Q involve different parameters, so they are maximised separately:

   - The update for π has a closed form, found using a Lagrange multiplier for the constraint $\sum_r \pi_{kr} = 1$:

$$\pi_{kr}^{(t+1)} = \frac{\sum_i w_{ik}^{(t)} C_{ir}}{\sum_i \sum_s w_{ik}^{(t)} C_{is}}.$$

   - The update for (α, β) has no closed form. It is a proportional odds fit in which each nodule appears once for every grade k, with weight $w_{ik}^{(t)}$, and it is

computed with `polr`.

3. **Stopping Criterion:** Stop when $\left|\ell(\Psi^{(t+1)}) - \ell(\Psi^{(t)})\right| < \epsilon$ for a small threshold ϵ, and take $\hat{\Psi} = \Psi^{(t+1)}$.

EM never decreases the observed log-likelihood, that is, $\ell(\Psi^{(t+1)}) \geq \ell(\Psi^{(t)})$, but it may converge to a local maximum rather than the global one. The result can therefore depend on $\Psi^{(0)}$.

Estimates The algorithm is run on the training set from six starting values, the weights implied by the median rating and five random perturbations of them. All six reach the same log-likelihood, -5190.172, so we find no sign of a local maximum. The fitted structural model is

$$\begin{aligned}\operatorname{logit} \hat{P}(Z_i \leq k \mid x_i) = \hat{\alpha}_k - \big(&1.682 \cdot \texttt{log_volume} + 0.471 \cdot \texttt{elongation} \\ &- 0.323 \cdot \texttt{flatness} - 0.583 \cdot \texttt{solidity} \\ &+ 0.010 \cdot \texttt{compactness}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.983, -0.494, 1.968, 4.024)$. Table 5 compares the odds ratios with the naive primary model, and Table 6 gives the estimated matrix $\hat{\pi}$.

Predictor	$\hat{\beta}$ (EM)	Odds Ratio (EM)	Odds Ratio (Naive)
`log_volume`	1.682	5.375	3.781
`elongation`	0.471	1.601	1.395
`flatness`	-0.323	0.724	0.841
`solidity`	-0.583	0.558	0.710
`compactness`	0.010	1.010	1.048

Table 5: EM Estimates of the Structural Model

Latent Grade	Rated 1	Rated 2	Rated 3	Rated 4	Rated 5
1	0.956	0.044	0.000	0.000	0.000
2	0.103	0.395	0.499	0.002	0.000
3	0.035	0.255	0.541	0.169	0.000
4	0.000	0.092	0.313	0.397	0.197
5	0.004	0.000	0.099	0.194	0.703

Table 6: Estimated Rating Probabilities $\hat{\pi}_{kr}$

Interpretation

- **Structural Model:** The directions of the four main effects match the naive model. Each of these odds ratios is further from 1 under EM, and size remains the largest effect. Compactness is again close to 1. EM does not provide standard errors directly, so these differences are not yet tested.
- **Latent Grades:** Latent grades 1, 3, 4 and 5 are most often rated as themselves, whereas latent grade 2 is rated 3 more often than 2. The latent grades should therefore be read as ordered groups rather than as exact matches to the rating scale.

4 Hypothesis Testing

To formally evaluate our models, we conduct a series of hypothesis tests at the $\alpha = 0.05$ level, all on the training set. All of them are likelihood ratio tests. For a full model M_1 and a restricted model $M_0 \subset M_1$, with maximised log-likelihoods $\hat{\ell}_1$ and $\hat{\ell}_0$, the test statistic is

$$\Lambda = 2\big(\hat{\ell}_1 - \hat{\ell}_0\big).$$

By the Neyman–Pearson lemma, the likelihood ratio test is most powerful when both hypotheses are simple. Our hypotheses are composite, so we rely on Wilks' theorem, under which Λ is approximately χ^2_r under H_0 for large samples, where r is the number of restrictions imposed by M_0. The statistics are computed before rounding, so the last digit may differ from the rounded log-likelihoods shown.

Finding $\hat{\ell}_1$ and $\hat{\ell}_0$ Each is the value of the log-likelihood at its maximum, so every test needs two fits, one over the full parameter space Θ and one over the subset Θ_0 allowed by H_0:

$$\hat{\ell}_1 = \max_{\theta \in \Theta} \ell(\theta), \qquad \hat{\ell}_0 = \max_{\theta \in \Theta_0} \ell(\theta).$$

Setting a coefficient to zero is the same as removing that predictor, and all the remaining parameters are re-estimated in the restricted fit.

For the naive models, the equations $\nabla \ell(\alpha, \beta) = 0$ have no closed-form solution, so ℓ is maximised iteratively with BFGS, a quasi-Newton method. Writing $\theta = (\alpha, \beta)$, each step is

$$\theta^{(t+1)} = \theta^{(t)} + s_t B_t \nabla \ell\big(\theta^{(t)}\big),$$

where s_t is a step size and B_t approximates the inverse of the negative Hessian, updated from the change in the gradient between steps. Since ℓ is concave in (α, β), this reaches the global maximum.

For the EM model, $\ell(\Psi)$ is maximised with the algorithm of the previous section, in which the update for (α, β) in each M-step also uses BFGS. We then evaluate ℓ at the converged parameters. This likelihood need not be concave, so a fit may stop at a local maximum. We therefore fit the full model from six starting values, and each drop-one model from two, keeping the largest log-likelihood. The naive and EM log-likelihoods are based on different data, the median grade and the individual ratings, so they are only compared within each family.

4.1 Overall Model Significance

This test evaluates whether a naive model as a whole explains the malignancy grade better than a model with only the thresholds.

- **Null Hypothesis (H_0):** All coefficients are zero, so the predictors have no association with the malignancy grade.
- **Alternative Hypothesis (H_1):** At least one $\beta_j \neq 0$.

The primary model has 5 coefficients and the secondary model has 10, so Λ is compared with the χ^2_5 and χ^2_{10} distributions respectively.

Calculation and Inference The thresholds-only model has $\hat{\ell}_0 = -2552.429$. For the primary model,

$$\Lambda = 2\big(-2155.134 - (-2552.429)\big) = 794.590,$$

and for the secondary model,

$$\Lambda = 2\big(-2108.642 - (-2552.429)\big) = 887.574.$$

Both p-values are strictly less than 0.001, so we reject the null hypothesis for both models. The predictors are jointly associated with the malignancy grade.

4.2 Individual Parameter Significance

While the overall test evaluates a model as a whole, a drop-one test evaluates the contribution of each predictor given the others.

- **Null Hypothesis (H_0):** $\beta_j = 0$. The j-th predictor has no association with the malignancy grade once the other predictors are included.
- **Alternative Hypothesis (H_1):** $\beta_j \neq 0$.

The restricted model is the full model refitted without the j-th predictor, so $r = 1$ and Λ is compared with the χ^2_1 distribution. The full models have $\hat{\ell}_1 = -2155.134$ (primary) and $\hat{\ell}_1 = -2108.642$ (secondary).

Predictor	Primary Model			Secondary Model		
	$\hat{\ell}_0$	Λ	p-value	$\hat{\ell}_0$	Λ	p-value
log_volume	-2397.217	484.165	< 0.001	-2286.774	356.264	< 0.001
elongation	-2173.628	36.988	< 0.001	-2125.440	33.596	< 0.001
flatness	-2160.434	10.600	0.001	-2109.324	1.363	0.243
solidity	-2169.223	28.176	< 0.001	-2111.322	5.360	0.021
compactness	-2155.448	0.628	0.428	-2108.643	0.002	0.967
spiculation	–	–	–	-2122.351	27.418	< 0.001
lobulation	–	–	–	-2109.614	1.944	0.163
margin	–	–	–	-2115.826	14.367	< 0.001
texture	–	–	–	-2109.075	0.866	0.352
sphericity	–	–	–	-2108.643	0.002	0.969

Table 7: Drop-One Likelihood Ratio Tests for the Naive Models

Inference As shown in Table 7, we reject the null hypothesis for log volume, elongation, flatness and solidity in the primary model, and log volume has by far the largest statistic. We fail to reject it for compactness. In the secondary model, we reject it for log volume, elongation, solidity, spiculation and margin. Flatness is no longer significant once the semantic ratings are included. These tests are not adjusted for multiple comparisons.

4.3 Joint Significance of the Semantic Ratings

To justify including the semantic ratings, we test their joint significance against the primary model, which is nested in the secondary model.

- **Null Hypothesis (H_0):** The coefficients of spiculation, lobulation, margin, texture and sphericity are all zero. The ratings provide no additional explanatory power beyond the geometric features.
- **Alternative Hypothesis (H_1):** At least one of these coefficients is non-zero.

The secondary model adds 5 coefficients, so $r = 5$.

Calculation and Inference Evaluating the statistic gives

$$\Lambda = 2\big(-2108.642 - (-2155.134)\big) = 92.984,$$

with a p-value strictly less than 0.001 under the χ^2_5 distribution. We therefore reject the null hypothesis. The semantic ratings improve the model, although this is partly circular because they come from the same radiologists who rate malignancy.

4.4 Significance of the Features in the EM Model

The same tests apply to the EM model, using the observed-data log-likelihood $\ell(\Psi)$ rather than the complete-data one. Each restricted model is refitted by EM. The full model has $\hat{\ell}_1 = -5190.172$.

Overall Significance

- **Null Hypothesis (H_0):** $\beta = 0$. The features are not informative about the latent grade beyond the readers' ratings, so $p_{ik} = p_k$ for every nodule.
- **Alternative Hypothesis (H_1):** At least one $\beta_j \neq 0$.

The restricted model has $\hat{\ell}_0 = -5606.396$, which gives

$$\Lambda = 2\big(-5190.172 - (-5606.396)\big) = 832.449.$$

With $r = 5$, the p-value is strictly less than 0.001, so we reject the null hypothesis. The features are informative about the latent grade.

Individual Parameter Significance We then test $H_0 \colon \beta_j = 0$ against $H_1 \colon \beta_j \neq 0$ for each feature, with $r = 1$.

Predictor	$\hat{\ell}_0$	Λ	**p-value**
`log_volume`	-5424.701	469.058	< 0.001
`elongation`	-5210.130	39.917	< 0.001
`flatness`	-5200.096	19.848	< 0.001
`solidity`	-5212.612	44.880	< 0.001
`compactness`	-5190.181	0.018	0.894

Table 8: Drop-One Likelihood Ratio Tests for the EM Model

Inference As shown in Table 8, we reject the null hypothesis for every feature except compactness. This agrees with the primary naive model: the same four features are significant, and log volume again has the largest statistic.

4.5 Proportional Odds Assumption

The models assume that each coefficient is the same at every threshold. We test this on the naive models, where each threshold has its own coefficient $\beta_j^{(k)}$ under the alternative.

- **Null Hypothesis (H_0):** $\beta_j^{(1)} = \beta_j^{(2)} = \beta_j^{(3)} = \beta_j^{(4)}$. The effect of the j-th predictor is the same at every threshold.
- **Alternative Hypothesis (H_1):** The effect differs for at least one threshold.

We use likelihood ratio tests. The restricted model is the proportional odds model, and the full model lets the coefficients differ by threshold. Freeing one predictor adds 3 parameters, so $r = 3$. Freeing all predictors gives $r = 15$ for the primary model and $r = 30$ for the secondary model.

Calculation and Inference Freeing all predictors in the primary model gives

$$\Lambda = 2\big(-2059.178 - (-2155.134)\big) = 191.913,$$

and in the secondary model it gives

$$\Lambda = 2\big(-1973.441 - (-2108.642)\big) = 270.402.$$

Both p-values are strictly less than 0.001, so we reject the proportional odds assumption for both models. Table 9 gives the test for each predictor.

Predictor	Primary Model		Secondary Model	
	Λ	p-value	Λ	p-value
log_volume	120.209	< 0.001	121.891	< 0.001
elongation	9.483	0.024	10.471	0.015
flatness	47.476	< 0.001	52.975	< 0.001
solidity	30.034	< 0.001	19.680	< 0.001
compactness	42.540	< 0.001	37.296	< 0.001
spiculation	–	–	7.963	0.047
lobulation	–	–	9.788	0.021
margin	–	–	73.527	< 0.001
texture	–	–	56.110	< 0.001
sphericity	–	–	4.843	0.184

Table 9: Likelihood Ratio Tests of the Proportional Odds Assumption ($r = 3$)

Every predictor except sphericity violates the assumption at the $\alpha = 0.05$ level. Table 10 shows how the odds ratios of the primary model change across thresholds when all predictors are free.

Predictor	$Y > 1$	$Y > 2$	$Y > 3$	$Y > 4$
log_volume	1.924	2.427	4.592	5.960
elongation	1.551	1.390	1.368	1.008
flatness	0.607	0.873	0.999	1.543
solidity	0.550	0.780	0.763	0.615
compactness	1.243	1.062	0.768	0.834

Table 10: Odds Ratios at Each Threshold, Primary Model

Size matters much more for separating the high grades than the low ones, so its single odds ratio is a summary rather than a constant multiplier. The effects of flatness and compactness change direction across thresholds, so their single odds ratios should not be interpreted on their own. The EM model assumes proportional odds for the latent grade, and we do not test this directly.

5 Conclusion

This study developed a multiple linear regression model to explain penguin body mass using flipper length, culmen length, culmen depth, and species. Motivated by allometric

scaling, a log transformation was applied to obtain a log-linear model that better captures the underlying relationships.

The fitted model demonstrated strong explanatory power, accounting for about 83.5% of the variation in log body mass. All predictors, except the intercept, were highly significant. Flipper length, culmen length, and culmen depth each showed a positive association with body mass, while the species indicators captured meaningful differences among penguin species.

The inferential results supported the usefulness of the model. The overall F-test confirmed statistical significance, and the joint test showed that species provides additional explanatory power beyond physical measurements. Confidence and prediction intervals were consistent with the fitted model.

Diagnostic analysis indicated that the regression assumptions are reasonably satisfied. Residual plots showed no major patterns, the Q-Q plot suggested approximate normality, and the chi-square test did not detect significant deviations. Overall, the model provides both strong predictive performance and a meaningful representation of the relationship between body mass and morphological traits.

A Appendix

A.1 Dataset

The dataset used in this study is the Palmer Penguins dataset[**?**], originally collected by Kristen B. Gorman and made publicly available by Allison Marie Horst, Alison Presmanes Hill, and Kristen B. Gorman.

It can be accessed through the following sources:

- GitHub: github.com/allisonhorst/palmerpenguins
- Kaggle: kaggle.com/datasets/parulpandey/palmer-archipelago-antarctica-penguin-data

A.2 Code

The code used in this project is divided based on functionality:

- R Code (Computation and Model Fitting): R Code Link
- Python Code (Visualization and Plotting): Python Code Link

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